A Mixed-Integer Linear Programming Approach for Optimal Workout Schedule Generation

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Abstract

1 Introduction

In the modern era, health and physical fitness have become central priorities for many individuals. However, the path to physical fitness is often cluttered with confusion (Cooper, 2022). A person walking into a typical gym is immediately confronted with a bewildering array of choices. There are racks of dumbbells, rows of complex machines, open areas for bodyweight movements, and various cables and pulleys. This project solves a weekly workout scheduling problem as a mixed-integer linear program (Melkonian, 2019). The goal is to automatically construct a 6-day split using a large exercise catalog, while enforcing realistic training constraints (time, volume caps, recovery spacing, frequency targets) and maximizing plan quality using exercise ratings as the utility signal.

2 Data Source

2.1 Variables

For this project, the source of information is the "Gym Exercise Dataset," also known as the "Mega Gym Dataset" available on Kaggle (Murarka, 2024). This dataset serves as a digital encyclopedia of human movement, providing a structured look at the chaotic environment of a fitness center (Murarka, 2024).

Each row in this dataset represents a unique exercise. In total, the dataset contains more than 2,900 distinct rows, with 9 primary columns as follows:

- *Title and Description*: The Title gives the exercise a name, such as "Barbell Bench Press" or "Banded Crunch". The Description provides a brief text explanation of the movement.

- *Type*: This variable categorizes the exercise based on its primary purpose, including 5 types - "Strength," "Cardio," "Stretching," "Plyometrics," and "Powerlifting". This column is critical for filtering. Since the goal of this project is to create a muscle-building routine, the computer must be able to distinguish between a "Strength" exercise and a "Cardio" exercise.

- *BodyPart*: The BodyPart column identifies which specific muscle group is the primary mover in the exercise, including "Abdominals," "Chest," "Quadriceps" (the front of the thigh), "Lats" (the large muscles of the back), "Triceps," and "Biceps". This information is vital for creating a balanced workout.

- *Equipment*: lists the tools required to perform the movement. This is a practical constraint which helps the user understand the logistical requirements of the workout.

- *Level*: a difficulty rating to the exercise, typically labeled as "Beginner," "Intermediate," or "Advanced".

- *Ratings (0–10)*: is the key quality signal in the optimization model: higher-rated exercises are treated as more efficient choices for driving hypertrophy/strength outcomes for their target muscle, so the solver prioritizes them when allocating limited weekly time and exercise slots. This is a defensible proxy because community ratings typically reflect observed results and execution practicality (how doable the movement is with correct form), so a high score (e.g., 9.5) indicates an exercise that users trust and repeatedly select. Exercises with low or missing ratings are treated as lower-confidence options and therefore contribute less (or are excluded) from the plan.

2.2 Data Cleaning and Preparation

Estimating Time: using The Logic of Sets and Repetitions Standard strength-training guidelines recommend performing each exercise in sets, where a set consists of several repetitions completed back-to-back (Transparent Labs, n.d.). Exercise science research generally suggests completing three to four sets per exercise, which provides enough training volume to stimulate muscle growth (Cooper, 2022) without exceeding typical workout durations. Each exercise therefore has two major time components: the work time, when the weight is actually being lifted, and the rest time, during which the muscles recover before the next set. While the lifting portion of a set may take only about 30 seconds, the rest period is considerably longer. Strength-training sources commonly recommend 60 seconds to several minutes of rest between sets, depending on the intensity of the exercise (EōS Fitness, 2025).

To create a consistent and computationally usable measure of time, the project applies a standardized time formula to every exercise in the dataset. Each exercise is assumed to require 3 sets, with a 45-second work duration per set (including small setup movements) and a 90-second rest period between sets. Using these assumptions, the total time for a single exercise becomes:

- Sets: Assumed to be 3 for every exercise.

- Work Duration: 45 seconds (including setup and the lifting itself).
- Rest Duration: 90 seconds (1.5 minutes) between sets.
- Total for one exercise: 405 seconds or 6.75 minutes.

For the purpose of the model, this value was likely rounded or adjusted slightly, but the core concept remains: every exercise costs roughly 6 to 7 minutes of the user's 90-minute budget. This estimation allows the computer to trade off "Rating" against "Time." If an exercise takes 7 minutes but has a rating of only 2.0, it is a bad investment. If it takes 7 minutes and has a rating of 9.5, it is a great investment.

2.3 Categorizing exercises

Another level of detail our project looks into is mechanics: whether an exercise is "Compound" (multi-joint) and "Isolated" (single-joint) exercises. Compound exercises, like squats, use many muscles at once (Transparent Labs, n.d.). Isolated exercises, like bicep curls, use only one.

Force: This categorizes exercises as "Push" (pushing away from the body) or "Pull" (pulling towards the center). This data is essential for the "Push-Pull-Legs" scheduler (Brama, 2023). The computer needs to know which exercises belong on "Push Day" and which belong on "Pull Day."

3 Main model

Objective Function: For the workout scheduler, we want the highest-quality weekly plan. Quality is defined by the exercise *Rating* provided in the dataset.

$$\max \sum_{d \in D} \sum_{e \in E} r_e \, x_{e,d}$$

Decision Variables:

$$x_{e,d} = \begin{cases} 1 & \text{if exercise } e \text{ is scheduled on day } d, \\ 0 & \text{otherwise} \end{cases} \qquad y_{m,d} = \begin{cases} 1 & \text{if muscle } m \text{ is trained on day } d, \\ 0 & \text{otherwise} \end{cases}$$

The Constraints: Several constraints are applied to ensure a realistic and physiologically meaningful training schedule.

Daily Time Capacity: The total estimated workout time on any given day may not exceed 90 minutes.

$$\sum_{e \in E} t_e \, x_{e,d} \;\leq\; T_{\max} \qquad \forall d \in D$$

Daily Volume Bounds: Each workout day must include between 5 and 8 exercises to ensure sufficient stimulus without excessive volume.

$$E_{\min} \;\leq\; \sum_{e \in E} x_{e,d} \;\leq\; E_{\max} \qquad \forall d \in D$$

Exercise–Muscle Linking: If an exercise targeting muscle m is performed on day d, that muscle must be marked as active.

$$y_{m,d} \;\geq\; x_{e,d} \qquad \forall e \in E, \forall d \in D \text{ where exercise } e \text{ trains } m$$

Major Muscle Frequency and Volume Limits: Major muscles (chest, lats, quadriceps, hamstrings, glutes, shoulders, biceps, triceps, lower back) must be trained on at least two distinct days per week. Additionally, no more than three exercises targeting the same major muscle may appear on a single day.

$$\sum_{d \in D} y_{m,d} \;\geq\; 2 \qquad \forall m \in M_{\text{major}}$$

$$\sum_{e \in E_m} x_{e,d} \ \leq \ 3 \qquad \forall m \in M_{\text{major}}, \forall d \in D$$

Minor Muscle Frequency and Volume Limits: Minor muscles must be trained on at least one day per week, with a maximum of two exercises per day targeting any given minor muscle.

$$\sum_{d \in D} y_{m,d} \ \geq \ 1 \qquad \forall m \in M_{\text{minor}}$$

$$\sum_{e \in E_m} x_{e,d} \ \leq \ 2 \qquad \forall m \in M_{\text{minor}}, \forall d \in D$$

Recovery Spacing: A muscle may not be trained on consecutive days, ensuring a minimum recovery period between sessions.

$$y_{m,d} + y_{m,d+1} \ \leq \ 1 \qquad \forall m \in M, \ d = 1, \ldots, 5$$

$$y_{m,6} + y_{m,1} \ \leq \ 1 \qquad \forall m \in M$$

Exercise Uniqueness: Each exercise may appear at most once per week, promoting variety in the schedule and preventing over-reliance on any single highly rated movement.

$$\sum_{d \in D} x_{e,d} \ \leq \ 1 \qquad \forall e \in E$$

4 Results

The output of the optimization model is a complete six-day workout schedule that follows the Push-Pull-Legs (PPL) split, a training structure widely cited in the exercise science literature as an effective method for organizing resistance training programs (Brama, 2023; Cooper, 2022). This approach categorizes exercises based on primary movement patterns and the muscle groups involved, allowing for systematic training and recovery across the week.

To interpret the results, it is necessary to understand the Push-Pull-Legs framework. The "Push" category includes muscles responsible for pushing weight away from the body, primarily the chest (pectorals), shoulders (deltoids), and triceps. Representative exercises within this category include the bench press, which involves pushing a bar away from the chest, and the overhead press, which requires pressing a load above the head. The "Pull" category consists of muscles that pull weight toward the body, including the back muscles (latissimus dorsi, trapezius, and rhomboids) and the biceps. Common examples of pull exercises are pull-ups, which involve pulling the body toward a fixed bar, and rowing movements that pull resistance toward the torso. The "Legs" category encompasses the entire lower body, including the quadriceps, hamstrings, gluteal muscles, and calves. Because the lower body involves a large amount of muscle mass and is typically subjected to high mechanical loading, it is often advised that leg-focused training be performed once per week to allow adequate recovery.

The schedule operates on a six-day cycle, commonly organized as Push, Pull, Legs, Push, Pull, and Rest. This structure allows each upper-body muscle group to be trained twice per week, a frequency that research suggests is optimal for promoting muscle hypertrophy (Cooper, 2022; EōS Fitness, 2025). To further enhance adherence and reduce monotony, the optimization model generated two variations for each training day, labeled A and B, providing structured variety while maintaining consistency within the overall training framework.

Table 1: Optimized Push Day A

Exercise	**Muscle Group**	**Reason for Selection**
Barbell Bench Press	Chest	A compound push movement with consistently high ratings. It serves as the fundamental chest builder.
Incline Dumbbell Press	Chest	Targets the upper chest specifically. Dumbbells provide a greater range of motion, complementing the barbell press.
Overhead Shoulder Press	Shoulders	The primary mass-building exercise for shoulders; also engages the triceps and core as stabilizers.
Lateral Raises	Shoulders	A targeted isolation movement for the side deltoids. Selected to satisfy the “two shoulder exercises” constraint.
Triceps Pushdown	Triceps	A highly rated isolation exercise that focuses purely on the triceps without fatiguing the shoulders.
Dumbbell Tricep Extension	Triceps	Adds additional triceps volume to ensure full muscular fatigue and complete the push-day profile.

This workout demonstrates the model's ability to prioritize ”Compound” movements (exercises that use multiple joints). The Bench Press and Overhead Press are placed first. This is standard fitness advice because these moves require the most energy. The isolation moves (Lateral Raises, Pushdowns) come later. The estimated time for these 6 exercises would be approximately 6 times 7 mins = 42 mins, well within the 90-minute limit.

Table 2: Optimized Pull Day A

Exercise	**Muscle Group**	**Reason for Selection**
Pull-ups	Lats (Back)	Often rated near 10/10 in fitness datasets. Provides significant back “width” through vertical pulling.
Barbell Deadlift	Lower Back	A powerhouse compound exercise. Although it uses the legs heavily, it is categorized as a back movement in many datasets. Excellent for strength development and caloric demand.
Bent Over Barbell Row	Middle Back	Builds mid-back “thickness.” Complements Pull-ups by providing horizontal pulling balance.
Barbell Curl	Biceps	The fundamental bicep builder. Simple, effective, and consistently highly rated.
Hammer Curls	Biceps/Forearms	Targets the brachialis and forearms. Adds movement variety and improves arm development from multiple angles.

The inclusion of the Deadlift is significant. The research snippets highlight the Deadlift as a major compound move. The model likely selected it because of its high rating in the ”Strength” category (Brama, 2023; Cooper, 2022). Placing it on Pull day is a common strategy to keep it away from Squats on Leg day, allowing the lower back to recover.

Table 3: Optimized Leg Day A

Exercise	**Muscle Group**	**Reason for Selection**
Barbell Squat	Quadriceps	Often referred to as the "King of Exercises." Consistently the highest-rated leg movement and essential for overall lower-body mass.
Leg Press	Quadriceps	Allows heavy loading with reduced spinal compression compared to squats, providing safe additional quad volume.
Romanian Deadlift	Hamstrings	Trains the posterior chain through hip hinging. Complements the knee-dominant squat for balanced leg development.
Lying Leg Curls	Hamstrings	An isolation exercise that fully fatigues the hamstrings after Romanian Deadlifts.
Standing Calf Raises	Calves	Ensures adequate lower-leg training and satisfies the requirement to include calf-focused work.

The balance here is mathematically perfect. We have two exercises for the front of the leg (Squat, Leg Press) and two for the back of the leg (RDL, Leg Curls). This prevents muscle imbalances that can lead to injury. The "Squat" is described in the snippets as a "competitive" and "classic" way to start leg day , confirming the model's choice to prioritize it.

One of the strengths of the generated schedule is that it includes "B" days (Push B, Pull B, Legs B). The research suggests that variety helps avoid strain and keeps the user engaged (Cooper, 2022; Transparent Labs, n.d.).The optimization model achieves this by re-running the calculation but forcing it to pick different exercises than the ones chosen for the A days.

- Push B: Instead of Barbell Bench Press, the model might select Dumbbell Bench

Press. This hits the chest differently. Instead of Tricep Pushdowns (cables), it might pick Skullcrushers (barbell).

- Pull B: Instead of Pull-ups, it might choose Lat Pulldowns (machine). This is beneficial for users who might be too tired for Pull-ups later in the week.
- Legs B: Instead of Back Squats, the model might select Front Squats or Lunges. Lunges are excellent for stability and ensure that one leg isn't stronger than the other.

5 Implementation and Interactive Application

To demonstrate the practical applicability of the optimization model, we implemented the framework as an interactive web application using Streamlit. The app allows users to generate customized weekly workout plans, directly leveraging the optimization logic described earlier, while abstracting the mathematical complexity behind the scenes. Through a user-friendly interface, the app provides flexibility in selecting various parameters, such as the desired focus (e.g., strength, hypertrophy), fitness level (e.g., beginner, intermediate, advanced), and equipment availability (e.g., home gym or full gym setup). These inputs enable users to generate optimized workout splits that respect time constraints, recovery rules, and muscle-group balance, all while ensuring that high-rated exercises are prioritized for maximum efficiency. Users can explore multiple training splits (Push-Pull-Legs or other variations) and adjust specific details of their workout schedule, making it highly customizable. This deployment demonstrates how prescriptive analytics can be translated into a real-world decision-support tool, empowering users to make data-driven decisions about their fitness routines and improving their adherence to a balanced, effective workout plan.

The application is publicly accessible at: https://da353-workoutoptimization.streamlit.app

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